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Original Article

The influence of different sitting postures on head/neck posture and muscle activity

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ABSTRACT

To date the influence that specific sitting posture has on the head/neck posture and cervico-thoracic muscle activity has been insufficiently investigated. Therefore the aim of this study was to investigate whether three different thoraco-lumbar sitting postures affect head/neck posture and cervico-thoracic muscle activity.

Twenty (10 men, 10 women) asymptomatic subjects were placed in 3 standardized thoraco-lumbar sitting postures (lumbo-pelvic, thoracic upright and slump) to investigate their influence on cervico-thoracic muscle activity and head/neck posture.

There were significant differences in lumbar and thoracic curvatures in the 3 different sitting postures ($P < 0.002$). Slump sitting was associated with greater head/neck flexion, anterior translation of the head ($P < 0.001$) and increased muscle activity of cervical erector spinae (CES) compared to thoracic and lumbo-pelvic sitting ($P = 0.001$). Thoracic upright sitting showed increased muscle activity of thoracic erector spinae (TES) compared to slump and lumbo-pelvic postures ($P = 0.015$). Upper trapezius (UT) demonstrated no significant difference in muscle activation in the 3 sitting postures ($P < 0.991$). This study demonstrates that different sitting postures affect head/neck posture and cervico-thoracic muscle activity. It highlights the potential importance of thoraco-lumbar spine postural adjustment when training head/neck posture.

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1. Introduction

Neck pain has been associated with a variety of occupations (Sommerich et al., 2000; Ariens et al., 2001). Development of neck symptoms in these occupations may be associated with adopting sustained non-neutral spinal postures with increased muscle activation of the neck-shoulder stabilizers resulting in higher levels of cervical spine loading (Szeto et al., 2005). In support of this, Falla et al. (2007a) reported that individuals with chronic neck pain demonstrate a reduced ability to maintain an upright sitting posture.

Previous evidence suggests that significant changes in trunk muscle activation occur as a result of changes in thoraco-lumbar sitting postures (Schuldt et al., 1986). O'Sullivan et al. (2002) reported a reduction in activity in the superficial lumbar multifidus, internal oblique and thoracic erector spinae (TES) in upright sitting when compared to slumped sitting. More recently O'Sullivan et al. (2006) compared the effect of three different thoraco-lumbar sitting postures (slump, thoracic upright and lumbo-pelvic) on spinal curvature and trunk muscle activation in a pain free cohort. These postures have been observed in clinical practice and reflect a spectrum of lumbo-pevic postures (O'Sullivan, 2004). The authors reported that the three sitting postures all resulted in different levels of trunk muscle activity. This was the first study to identify and measure the difference in kinematics and muscle activity between two upright sitting postures (thoracic upright and lumbo-pelvic upright) and slump sitting in a pain-free population, and the results highlight the influence that specific spinal postures have on trunk muscle activity.

Evidence to link the influences of thoraco-lumbar spine posture to cervical posture and cervico-thoracic muscle activity is limited. Falla et al. (2007b) compared activation of the deep neck flexor

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(DNF) muscles, TES and lumbar multifidus during independent and facilitated correction of sitting from a slouched posture to a neutral lumbo-pelvic posture in 10 female subjects with chronic neck pain. The results showed that activity of the DNF muscles and lumbar multifidus were significantly greater when postural correction was facilitated by a therapist. However, no spinal kinematics measures were taken to quantify differences in both cervical and thoraco-lumbar posture.

Using surface EMG, McLean (2005) investigated muscle activation in the cervico-brachial region in different sitting postures. The findings of this study revealed reduced cervico-thoracic muscle activation (including cervical erector spinae and upper trapezius), when subjects adopted an upright sitting posture. However sitting posture was determined by using plumb lines to set body position, therefore kinematics were not quantified, and the effects of different sitting postures on the cervical spine posture were not investigated.

Black et al. (1996) used a three dimensional digitizer system and a hand held inclinometer to measure changes in cervical and lumbar posture in different sitting postures. The authors highlighted the kinematic link between lumbo-pelvic sitting and cervical spine posture. However, no investigation of cervico-thoracic muscle activity was performed.

To date no study has specifically investigated whether different thoraco-lumbar sitting postures influence head/neck posture and cervico-thoracic muscle activity. Therefore, the aim of this study was to investigate whether the three different sitting postures described by O'Sullivan et al. (2006) (lumbo-pelvic, thoracic upright and slump) affect head/neck posture and cervico-thoracic muscle activity.

2. Methods

2.1. Subjects

Twenty asymptomatic subjects aged between 18 and 35 years participated in this study. Ten males (mean ± SD height; 183.2 ± 9.2 cm; mass, 81.0 ± 2.9 kg; age, 25.7 ± 3.5 yrs) and ten females (mean ± SD height; 166.5 ± 5.6 cm; mass, 62.5 ± 9.5 kg; age, 24.5 ± 4.2 yrs) were recruited. Subjects had no history of back or neck pain within the last two years and were excluded if they currently experienced dizziness, fainting, symptomatic irregular heartbeat, or if they participated in activities known to strengthen the neck more than daily living activities (adapted from Sommerich and colleagues, 2000). Ethical approval was obtained from the institutional Human Research Ethics committee at Curtin University of Technology and informed consent was obtained from all subjects prior to testing.

2.2. Experimental protocol

This study utilized a within-subjects design with subjects being positioned into three thoraco-lumbar sitting postures as defined by O'Sullivan et al. (2006):

- Slump Sitting – posterior rotation of the pelvis, thoraco-lumbar spine relaxed while looking straight ahead (Fig. 1a).
- Lumbo-pelvic Upright Sitting – anterior rotation of the pelvis in order to achieve a neutral lordosis of the lumbar spine and relaxation of the thorax (Fig. 1b).
- Thoracic Upright Sitting – anterior rotation of the pelvis, thoraco-lumbar spine extended and with shoulder blades slightly retracted (Fig. 1c).

Each trial was conducted on an adjustable stool with subject's hips and knees at 90 degrees, feet positioned shoulder width apart with arms relaxed at the side of their body. Subjects were requested to view a designated point ahead at eye level. Prior to data collection subjects were positioned by two qualified physiotherapists in the three sitting postures in random order. Each subject completed three trials of 5 s for each sitting posture. Between trials subjects remained seated in order to be repositioned by the physiotherapists. Once the next sitting posture was achieved the trial was conducted.

2.3. Data collection

While maintaining the above mentioned sitting postures cervico-thoracic muscle activation (surface EMG) and kinematics of the thoraco-lumbar and head/neck posture (electromagnetic tracking device) were synchronously recorded. The posture of the thoraco-lumbar spine was collected during the trial to confirm that the subjects were in the desired thoraco-lumbar posture. The thoraco-lumbar posture was quantified via the thoracic angle (C7 relative to T12) and lumbar angle (T12 relative to S2) (Burnett et al., 2009). EMG data were collected bilaterally from the following muscles:

- Cervical Erector Spinae (CES) – 2 cm lateral from the C4 spinous process (Strimpakos et al., 2005; Netto and Burnett, 2006a).
- Upper Trapezius (UT) – lateral to the half-way point of an imaginary line formed by the posterior aspect of the acromion and the spinous process of C7 (Hermens et al., 2000).
- Thoracic Erector Spinae at T4 level (TES T4) – 5 cm lateral from the spinous process at T4, adapted from Solomonow et al. (2003).

EMG data was collected using an eight channel Octopus Cable Telemetric system (Bortec Electronics Inc., Calgary, Canada) operating at 1000 Hz (bandwidth 10–500 Hz, common mode refection ratio > 115 dB at 60 Hz). All raw EMG signals were amplified with an overall gain of 2000 and data were recorded using a customized software program (LabVIEW V7.0, National Instrument, Austin, TX). Subject's skin was shaved, abraded and cleaned with alcohol to reduce skin impedance to below 5 kΩ. 12 mm diameter Ag/AgCl disposable surface electrodes (Uni-Patch, Wasbasha, MN) were taped to the skin in a bipolar configuration. An inter-electrode distance of 20 mm was chosen for all sites, as recommended to avoid unstable recordings by not exceeding ¼ of the fiber length of the muscle (Hermens et al., 2000). sEMG channels shared the same ground placement reference of the clavicle.

For the purpose of EMG data normalization, a series of 5 s maximum voluntary isometric contractions (MVIC's) of the abovementioned muscles was performed prior to the sitting trial. Full details of these normalization procedures have been outlined previously (Netto and Burnett, 2006a).

To determine the influence of different sitting postures on the cervical spine and to quantify the kinematics of the thoraco-lumbar spine in the seated postures an electromagnetic tracking system operating at 25 Hz was used (3Space Fastrak TM, Polhemus Navigation Science Division, Kaiser Aerospace, Vermont). This system has shown to be both valid and reliable for these purposes (Pearcy and Hindle, 1989; O'Sullivan et al., 2006).

To quantify head/neck posture, sensors were placed on the main protuberance of the forehead and the spinous process of C7 allowing head/neck flexion in sagittal plane and head translation/displacement in horizontal plan to be recorded (Dall'Alba et al., 2001; Netto and Burnett, 2006b).

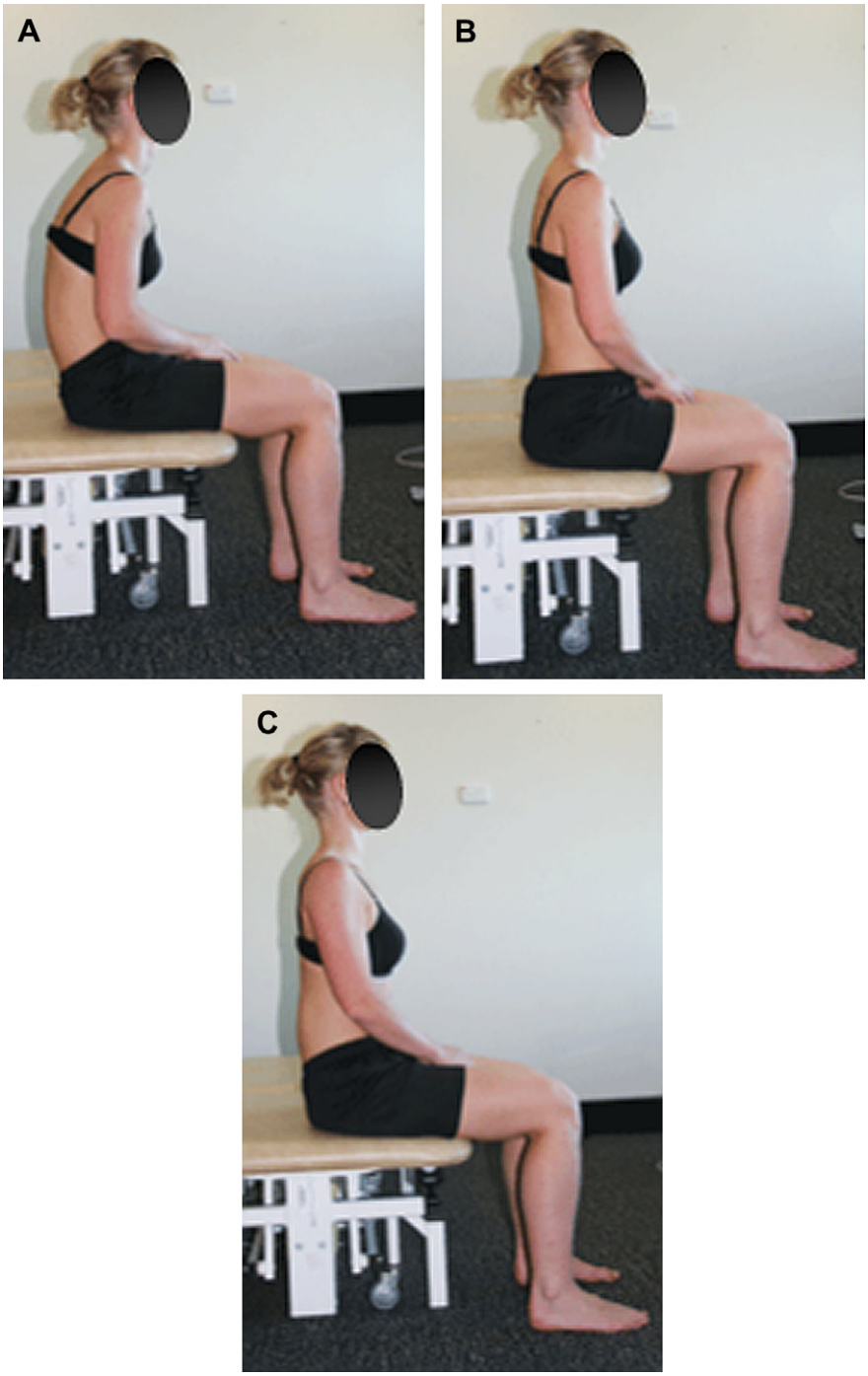


Fig. 1. Thoraco-lumbar sitting postures: (a) Slump sitting, (b) Lumbo-pelvic sitting, (c) Thoracic upright sitting.

2.4. Data management

Raw sEMG signals were demeaned, full wave rectified and low pass filtered at 4 Hz using a fourth order dual pass Butterworth digital filter to form a linear envelope. Heartbeat artefact was minimised in the cervico-thoracic muscles by using a fourth order Butterworth high pass filter with a cut-off frequency of 30 Hz (Drake and Callaghan, 2006). EMG data was normalized to the MVIC values obtained for each muscle. MVIC values were obtained from the average of the last two of the three maximal contractions and a 200 ms moving window was applied to the linear envelope (Netto and Burnett, 2006a; Burnett et al., 2007).

Head rotation relative to C7 (head/neck flexion), head translation (head relative to C7), thoracic angle and lumbar angle were determined from raw spinal kinematic data. Raw data from the Fastrak were output in a lateral bending (Z), flexion/extension (Y) and axial rotation (X), Cardan angle sequence and matrix algebra procedures (Burnett et al., 2009) transformed the data to a YZX order of rotations (Hof et al., 2001).

The inter-trial reliability for the muscle activation and spinal kinematics data were assessed using the intra-class correlation coefficient (ICC) and the relative standard error of measurement (%SEM) (Norton et al., 2000). Due to good reliability (Tables 1 and 2) of these data the three trials were averaged and paired *t*-tests

Table 1
Reliability indices for cervico-thoracic electromyography data.

	Slump		Thoracic		Lumbo-pelvic	
	ICC	%SEM	ICC	%SEM	ICC	%SEM
LCES	0.9	1.5	0.9	1.3	0.9	1.3
RCES	0.9	1.4	0.9	1.3	0.9	2.1
LUT	0.9	0.9	0.9	1.3	0.7	2.2
RUT	0.9	0.9	0.9	0.7	0.8	1.4
LTES	0.7	2.0	0.7	1.7	0.8	0.8
RTES	0.8	1.4	0.7	1.6	0.8	1.0

Intra-class correlation coefficient (ICC) and relative standard error of measurement (%SEM) data are presented. LCES represents left cervical erector spinae; RCES represents right cervical erector spinae; LUT represents left upper trapezius; RUT represents right upper trapezius; LTES represents left thoracic erector spinae; RTES represents right thoracic spinae.

revealed no significant difference between sides for the sEMG data, therefore left and right side data were also averaged.

2.5. Statistical analyses

One-way ANOVA with repeated measures was used to detect differences between the three sitting postures in lumbar and thoracic curvatures, head/neck flexion relative to C7 and head translation relative to C7 and cervico-thoracic muscle activation. Within-subject contrasts were used to detect further differences between muscle activity as well as kinematics within each sitting posture. All statistical analysis was performed using the Statistical Package for the Social Sciences (SPSS) statistical analysis software V13.0 (SPSS, Chicago, IL).

3. Results

3.1. Thoraco-lumbar postures

The kinematics for the lumbar and thoracic spinal curvatures in the three sitting postures are presented in Fig. 2. There were significant differences evident in lumbar and thoracic curvature measured for the three different sitting postures ($P < 0.002$), indicating that they represented the thoraco-lumbar postures as defined by O'Sullivan et al. (2006). Post-hoc testing revealed the following:

- Thoracic upright sitting posture when compared to lumbo-pelvic sitting posture was associated with significantly less thoracic flexion ($P < 0.001$) and lumbar extension ($P < 0.001$).
- Thoracic upright sitting posture when compared to slump sitting posture was associated with significantly less thoracic flexion ($P < 0.001$) and greater lumbar extension ($P < 0.001$).

Table 2
Reliability indices for spinal curvatures kinematics.

	Slump		Thoracic		Lumbo-pelvic	
	ICC	%SEM	ICC	%SEM	ICC	%SEM
Cx angle	0.9	5.4	0.9	4.3	0.9	5.5
Cx AP	0.8	0.7	0.9	4.4	0.8	0.7
Tx angle	0.9	2.9	0.9	3.4	0.9	4.2
Lx angle	0.9	1.6	0.9	0.6	0.8	3.3

Intra-class correlation coefficient (ICC) a relative standard error of measurement (%SEM) data are presented. Cx angle stands for head/neck rotation in sagittal plane; Cx AP represents head translation in transversal plane; Tx angle represents thoracic rotation in sagittal plane and Lx-angle angle lumbar rotation in sagittal plan.

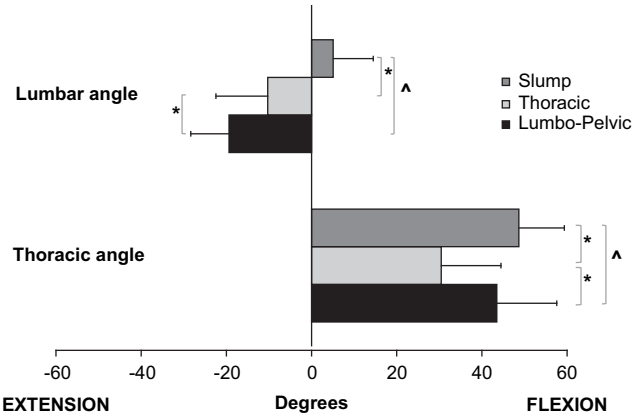


Fig. 2. Comparison of mean thoracic and lumbar curvatures across three sitting postures: slump, thoracic upright and lumbo-pelvic upright sitting. Error bars indicate SD, * indicates $p < 0.001$, and (^) indicates $p < 0.002$.

- Slump sitting posture when compared to lumbo-pelvic sitting was associated with significantly greater thoracic flexion ($P < 0.002$) and lumbar flexion ($P < 0.002$).

3.2. Head/neck kinematics

There were differences in head/neck posture as evidenced by data for sagittal plane flexion (Fig. 3) and horizontal translation (Fig. 4) in all three sitting positions. Specifically, the results from the head/neck kinematics included:

- Lumbo-pelvic sitting when compared to thoracic upright sitting was associated with significantly greater head/neck flexion relative to C7 ($P = 0.006$) and a trend towards greater anterior translation ($P = 0.064$).
- Slump sitting when compared to thoracic upright sitting demonstrated significantly greater head/neck flexion ($P < 0.001$) and anterior translation ($P < 0.001$).
- Slump sitting when compared to lumbo-pelvic sitting showed a significant greater head/neck flexion ($P < 0.001$) and anterior translation ($P < 0.001$).

3.3. Cervico-thoracic muscle activation

The normalized (%MVIC) cervico-thoracic muscle activity is illustrated in Fig. 5. The three sitting postures were associated with

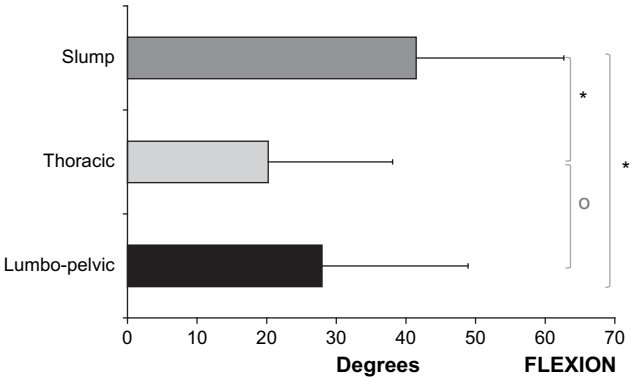


Fig. 3. Comparison of mean head/neck flexion relative to C7 across three sitting postures: slump, thoracic upright and lumbo-pelvic upright sitting. Error bars indicate SD, (*) indicates $p < 0.001$, and (**o**) indicates $p < 0.006$.

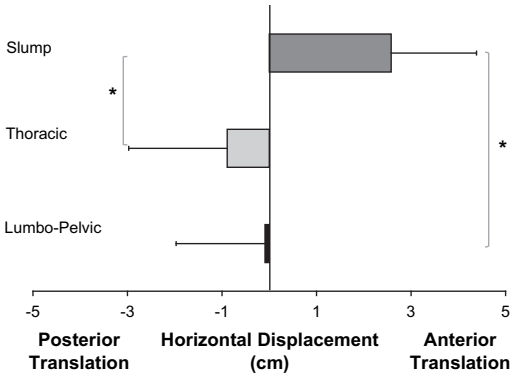


Fig. 4. Comparison of the mean horizontal head translation relative to C7 in relation to three sitting postures: slump, thoracic upright and lumbo-pelvic upright sitting. Error bars indicate SD, and (*) indicates $p < 0.001$.

a significant difference in the cervical erector spinae ($P < 0.033$) and thoracic erector spinae (TES T4) ($P < 0.015$). There was no significant difference evident for UT muscle activation in the three sitting postures ($P < 0.991$). The results from the study show:

- Thoracic upright sitting when compared to lumbo-pelvic sitting was associated with significantly less muscle activity in CES ($P = 0.033$) and significantly greater muscle activity in TES T4 ($P < 0.001$).
- Slump sitting when compared to thoracic upright sitting demonstrated significantly greater muscle activity in CES ($P = 0.001$) and less muscle activity in TES T4 ($P = 0.015$).
- Slump sitting when compared to lumbo-pelvic sitting was associated with significantly greater muscle activity in CES ($P = 0.001$) and TES T4 ($P = 0.015$).

4. Discussion

This study investigated whether different thoraco-lumbar sitting postures have an effect on cervico-thoracic muscle activity and posture. Previous research in this area is limited due to the fact that either the different sitting postures have not been directly quantified (Schuldt et al., 1986; McLean, 2005; Falla et al., 2007b) or cervical extensor muscle activity has not been measured (Black et al., 1996; Falla et al., 2007b).

This study showed that all three defined sitting thoraco-lumbar postures (slump, thoracic upright and lumbo-pelvic sitting) resulted in various differences in head/neck kinematics and muscle activation of CES and TES. These results demonstrate a clear link between thoraco-lumbar spine posture and head/neck posture and motor activity.

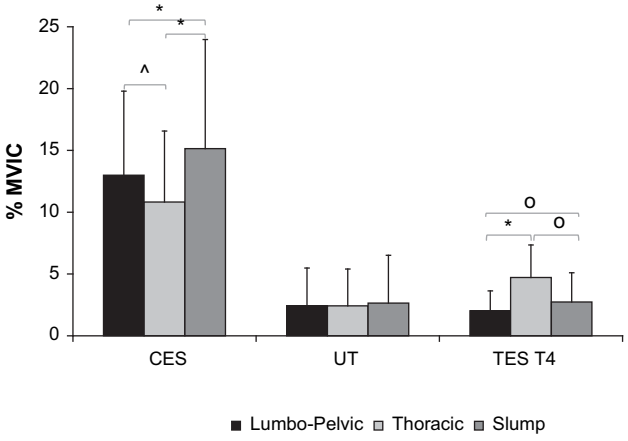


Fig. 5. Comparison of the mean of muscle activity (% of MVIC) in 3 muscle groups across 3 sitting postures: slump, thoracic upright and lumbo-pelvic upright sitting. Error bars indicate SD, (*) indicates $p < 0.001$, (**o**) indicates $p < 0.0015$, and (^) indicates $p < 0.0033$. CES indicates cervical erector spinae, UTR (upper trapezius), TES T4(thoracic erector spinae).

4.1. Slump sitting posture

When compared to lumbo-pelvic upright sitting, slump sitting was found to be associated with increased thoracic flexion and head/neck flexion with a greater anterior translation of the head. These kinematic findings were also associated with greater activation of the CES and TES T4 muscles. The increase in cervico-thoracic muscle activity in slump sitting compared to lumbo-pelvic sitting is presumably associated with greater anterior translation of the head in the slumped posture increasing the moment arm of the head and neck with respect to the instantaneous axis of rotation which would be located in the cervical spine. Presumably the CES and TES T4 would therefore need to increase their activity in order to support the weight of the head/neck in this forward head posture. Considering the multi-segmental attachments and greater activity of these muscles it could be speculated that higher levels of compressive loading would occur in the cervico-thoracic region in slump sitting. Previous studies report that slump sitting also results in increased upper cervical spine extension and lower cervical spine flexion (Black et al., 1996). This combined with increased head/neck flexion and neck extensor muscle activity may create focus stress regions in the cervical spine resulting in postural pain.

4.2. Thoracic upright sitting

Thoracic upright sitting when compared to lumbo-pelvic sitting was associated with increased thoracic extension and decreased head/neck flexion with a trend towards posterior translation of the head. Not surprisingly this was coupled with greater muscle activity of the TES T4 compared to the other postures, but in contrast to lumbo-pelvic and slump sitting a reduction in muscle activation of the CES was also noted. There are a number of possible causes for these findings. Firstly, as reported by O'Sullivan et al. (2006), the thoracic upright posture is achieved via thoracic spinal extensors rather than lumbo-pelvic spinal extensors. This was described as a substitution pattern where the activation of the TES T9 resulted in a relaxation of the lumbar multifidus (O'Sullivan et al., 2006). The same phenomenon may be the case in the cervical spine where the activation of the long cervico-thoracic extensors results in relaxation of the local cervical spine extensors. Secondly, the reduced activation of the CES in this posture may be due to increased activation of the superficial anterior neck muscles, resulting in reciprocal inhibition of the cervical extensors. Finally, in this posture there is no flexion moment of the head as it is retracted over the thorax, resulting in no requirement for activation of the cervical extensor muscles. Whatever the case, the overall increase in activation of the TES T4 in thoracic upright sitting may expose the cervico-thoracic spine to a greater compressive loading via their multisegmental attachment.

4.3. Lumbo-pelvic sitting

The lumbo-pelvic posture revealed relatively neutral head/neck alignment and reduced superficial extensor muscle activity of the cervico-thoracic region, when compared to slump sitting. These findings are consistent with recent research which demonstrated

increased deep neck flexor activity when neutral lumbo-pelvic posture was facilitated, compared to slumped and non-facilitated upright postures (Falla et al., 2007a,b). O'Sullivan et al. (2006) proposed that lumbo-pelvic sitting may be favourable for the lumbar and thoracic region as this position does not involve end range postures, and preferentially activates the lumbo-pelvic postural muscles without dominant activation of the TES and upper abdominal wall.

The only muscle not to change its activity significantly between the 3 postures was the UT. Johnston et al. (2008), described that UT did not relax after a copy-typing computer task in workers with neck pain compared with controls and workers without pain. However once the task was completed the control group had the UT muscle back to its baseline levels.

Szeto et al. (2009), compared cervico-thoracic muscle activity of female office workers with chronic neck pain and asymptomatic subjects in two resting positions (hands on laps and hands on keyboard). An increase in UT activity levels was reported when workers with neck pain had their hands resting on the keyboard. Conversely, no difference in UT activity levels between the two resting postures was found on the asymptomatic group.

Laursen et al. (2002) reports that head movement and eye–hand coordination tasks encourage UT muscle activity. However, Burnett et al. (2009) when assessing the flexion relaxation phenomena of cervico-thoracic muscles of 20 asymptomatic subjects in lumbo-pelvic sitting did not find UT myoeletric activity during head flexion and extension.

The findings of the current study suggest that UT's role does not vary in maintaining these static thoraco-lumbar sitting postures where the upper limb and eye–hand coordination demand is minimal.

4.4. Clinical relevance

Inspite of the limitations of the study as mentioned below, we consider that the results of this study may have relevance to the clinical setting.

Recent research by Straker et al. (2009) did not demonstrate a relationship between neck pain and posture in adolescents, although they reported that the lack of finding may reflect a lack of subject sub-grouping. Similarly, Szeto et al. (2002) revealed only a trend towards increased head tilt and neck flexion in symptomatic subjects when compared to asymptomatic subjects during a computer display task. However other research supports a relationship between head/neck posture and pain, with Falla et al. (2007b) demonstrating that symptomatic neck subjects were found to have a more forward head posture during a 10-minute computer task. Similarly, Yip et al. (2008) reported that patients with neck pain had more forward head postures when compared to asymptomatic individuals. Thoraco-lumbar posture was not assessed in these studies. The discrepancy in findings may relate to different clinical subgroups recruited to the various studies, a concept highlighted by Dankaerts et al. (2006) who only found a relationship between posture and LBP when sub-grouped based on pain provocation behaviours.

To date investigating the relationship between neck pain and posture has not to our knowledge been investigated in relation to sub-grouping neck pain. The findings of the current study may have implications for different patterns of cervical spine loading in different sitting positions. Further research is required to determine whether a relationship between these subgroups of spinal postures and neck pain exists. Establishing a link between neck pain and altered postural patterns in sitting would provide indications for postural re-education programs in the management of patients with postural related neck pain.

In the case of postural related cervico-thoracic pain, where pain is provoked with either protracted head posture, or retracted head position, attention to adjusting thoraco-lumbar posture may be important in order to alter patterns of tissue load and pain sensitivity on the cervico-thoracic region, rather than directing management at the cervical spine in isolation. These findings support clinical assertions, in relation to postural re-education (Jull et al., 2002) and other research (Falla et al., 2007b), that places an importance on the thoraco-lumbar posture in relation to changing head neck posture in neck pain disorders.

4.5. Limitations

There were some limitations of the study which need to be acknowledged. Firstly, we did not differentiate between the upper and lower cervical spine kinematics and the deep and superficial cervical flexors and sub-occipitals muscles were not measured in this study. The personal postural habits of the participants were not been assessed in this study as the aim was to assess the influence of three pre-defined thoraco-lumbar postures on head/neck posture and cervico-thoracic muscle activity. While the authors acknowledge that individual differences such as gender, body size and spinal curvature may have the potential to influence the kinematics and muscle patterning of the head/neck region, clear differences were observed in this region based on changes in thoraco-lumbar posture. Furthermore, we recruited subjects of both genders with similar body composition therefore, we acknowledge that the findings of this study may differ for older subjects and those with higher levels of BMI. Finally, given that this study investigated asymptomatic subjects over a short duration (5 s window), it is not known whether similar responses would be detected in symptomatic subjects or over longer time periods. This will be a focus of future studies.

5. Conclusion

Different upright sitting postures (lumbo-pelvic, thoracic upright and slump) result in altered muscle activity of the cervico-thoracic region, as well as head/neck kinematics. These findings demonstrate a clear link between thoraco-lumbar postures in sitting and head/neck posture and motor activity. These findings support important role of the thoraco-lumbar spine in postural training of the head and neck region.

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